Deep Learning Architectures for Offshore Production Monitoring: A Comparative Benchmark on Fault Detection and Gas Forecasting

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***Abstract*—Digital monitoring of offshore oil and gas operations demands models capable of detecting equipment faults and forecasting production volumes under challenging conditions: high class imbalance, limited labeled data, multi-well heterogeneity, and long forecast horizons. We present a systematic benchmark of eleven architectures—eight trained models (Bidirectional LSTM, DeepONet, Random Forest, PatchTST, TCN, FKM-AD, MambaSL, ConvTimeNet) and three foundation models (Chronos, TimesFM, TiRex)—evaluated on two real-world offshore datasets using a nested evaluation protocol with held-out test sets and inner cross-validation. For fault classification on the Petrobras 3W dataset (208,973 windows, 10 classes), a statistical feature extraction pipeline compresses 720-step windows into 14-descriptor feature matrices; DeepONet achieves the best held-out accuracy of 96.85% (macro F_1=0.964), while all trained models operating on features reach >96% after hyperparameter optimization, including Random Forest at 96.58%—demonstrating that the feature extraction pipeline captures discriminative structure accessible even to non-sequential learners. FKM-AD (Fourier-KAN Mamba Anomaly Detector) provides a dual-path evaluation: 96.7% on features vs. 94.9% on raw windows, quantifying the feature extraction benefit for state-space models. MambaSL [13] extends the dual-path analysis with a single-layer time-variant SSM: the feature path achieves 96.6% accuracy while the raw-window path reaches 91.9%—a 4.7 pp gap that exceeds FKM-AD's 1.8 pp gap, suggesting MambaSL's architecture benefits more from pre-extracted features. ConvTimeNet [14] contributes a pure-convolution perspective—deformable patch embedding plus structural re-parameterization—achieving 96.6% accuracy on features vs. 15.8% on raw windows, a dramatic 80.8 pp gap that represents the largest feature extraction benefit among all dual-path models, extending the inductive bias analysis to architectures that rely neither on attention nor on state-space mechanisms. TiRex provides a competitive zero-shot baseline (91.2%). For gas production forecasting on the Ganymede dataset (31,359 samples, 7 wells), foundation models dominate: TimesFM achieves R^2_{prod}=0.83 at the 7-day horizon, exceeding the best trained model (PatchTST, 0.56) by 48%. Statistical significance is confirmed via Friedman tests on both tasks ($p < 0.05$), with eleven model configurations in the 3W analysis (χ^2=43.5, $p < 0.001$, Nemenyi CD=6.75).**

***Index Terms*—offshore production monitoring, fault classification, time-series forecasting, deep learning, foundation models, benchmark**

I. Introduction

Offshore oil and gas platforms generate massive volumes of multivariate sensor data that encode early signatures of equipment failures, well integrity degradation, and production anomalies. Timely detection of these events is critical: undetected faults can lead to unplanned shutdowns costing millions of dollars per day and posing environmental and safety risks [1].

Traditional monitoring relies on physics-based models and manual threshold alarms, which struggle with the complexity of modern subsea systems. Deep learning offers a data-driven alternative, but the offshore domain presents unique challenges: (i) extreme class imbalance in fault data, where normal operation vastly outnumbers rare anomaly types; (ii) multi-well heterogeneity, where models trained on one well may not generalize to others; (iii) long-horizon forecasting requirements for production planning; and (iv) limited availability of labeled anomaly data.

Recent advances in foundation models (FMs) for time series [6]–[8] offer zero-shot capabilities without task-specific training, potentially addressing the labeled-data bottleneck. However, their performance on domain-specific offshore data relative to supervised approaches remains underexplored.

A. Contributions

We make the following contributions:

1) A **systematic benchmark** of eleven architectures—eight trained (LSTM, DeepONet, Random Forest, PatchTST, TCN, FKM-AD, MambaSL, ConvTimeNet) and three foundation models (Chronos, TimesFM, TiRex)—across two real-world offshore datasets covering fault classification and production forecasting.
2) A **statistical feature extraction pipeline** that compresses raw 720-timestep sensor windows into compact 14-descriptor matrices, reducing computation by ∼50× while preserving discriminative information.
3) A **nested evaluation protocol** with held-out test sets and inner cross-validation that provides both unbiased performance estimates and variance diagnostics, following best practices for ML evaluation [10].

4) A **multi-horizon analysis** comparing trained models and foundation models across 7, 14, 30, and 90-day forecasting horizons with per-well granularity.

II. Related Work

A. Fault Detection in Oil and Gas

The 3W dataset [1] introduced a realistic benchmark for rare event detection in offshore wells, containing multivariate sensor readings across 10 fault classes from Petrobras operations. Prior work has applied classical machine learning (Random Forests, SVMs) with hand-crafted features [2], and more recently, deep learning approaches including CNNs and LSTMs [3]. However, most studies use simple train/test splits or K-fold cross-validation without proper group separation, risking data leakage when multiple windows from the same well instance appear in both training and validation sets.

B. Production Forecasting

Gas and oil production forecasting has been addressed with physics-informed models (decline curve analysis), statistical methods (ARIMA, VAR), and increasingly deep learning [9]. Transformer-based architectures have shown strong performance on long-horizon time-series forecasting [5], while operator-learning frameworks like DeepONet [4] offer physics-compatible function-to-function mappings. Multi-well forecasting remains particularly challenging due to heterogeneous geological conditions and operational histories across wells.

C. Foundation Models for Time Series

The recent emergence of FMs pre-trained on massive time-series corpora has introduced zero-shot forecasting capabilities. Chronos [6] tokenizes time series into bins for language-model-style generation; TimesFM [7] uses a patched-decoder architecture trained on 100B timepoints; and TiRex [8] leverages xLSTM backbones for universal time-series representation. These models eliminate the need for task-specific training but may lack domain adaptation for specialized industrial data.

III. Datasets

A. 3W Dataset (Fault Classification)

The 3W dataset v2.0.0 [1] contains multivariate sensor recordings from 1,568 offshore well instances operated by Petrobras. We extract 208,973 sliding windows of 720 timesteps $\times$ 27 sensors, labeled across 10 classes: normal operation (class 0) plus 9 anomaly types including abrupt BSW increase, incipient BSW increase, spurious closure, severe slugging, flow instability, rapid productivity loss, quick restriction, slow restriction, and hydrate formation.

Table I summarizes the class distribution, which exhibits significant imbalance: class 2 (incipient BSW increase) contains only 1,986 samples (1.0%) while class 5 (flow instability) contains 36,323 (17.4%).

TABLE I: 3W dataset class distribution.

ID	Fault Type	Count	%
0	Normal operation	32,773	15.7
1	Abrupt BSW increase	25,091	12.0
2	Incipient BSW increase	1,986	1.0
3	Spurious closure	13,597	6.5
4	Severe slugging	9,614	4.6
5	Flow instability	36,323	17.4
6	Rapid productivity loss	15,905	7.6
7	Quick restriction	28,504	13.6
8	Slow restriction	19,179	9.2
9	Hydrate formation	26,001	12.4
	Total	**208,973**	**100.0**

B. Ganymede Dataset (Production Forecasting)

The Ganymede dataset contains daily gas production records from 7 wells on the Norwegian Continental Shelf, spanning periods of 1 to 14 years. After shutdown filtering (removing windows where the target is entirely zero, 35% of samples), the dataset contains 31,359 samples with 63 features per timestep (9 raw measurements plus EMA lag features). We construct input windows of 90 timesteps and evaluate multi-step forecasting at horizons $h \in \{7, 14, 30, 90\}$ days. Target values are z-score normalized using statistics computed from non-zero training samples only, preserving the distinction between productive and shutdown periods.

IV. Methods

A. Feature Extraction (3W)

Rather than feeding raw 720-step windows directly to models, we compress each window into a statistical feature matrix of shape (14×27)—14 descriptors computed independently per sensor channel. This approach follows established practice in the 3W literature and provides several advantages: reduced computation, invariance to minor temporal shifts, and explicit capture of distributional properties.

The 14 statistical features are:

1) **Mean** (μ): Central tendency of the sensor signal
2) **Standard deviation** (σ): Signal variability
3) **Minimum / Maximum**: Range endpoints
4) **Median**: Robust central tendency
5) **Skewness**: Distributional asymmetry
6) **Kurtosis**: Tail heaviness
7) **Linear slope**: Trend direction via least-squares regression
8) **Mean absolute change**: First-order roughness
9) **Count above mean**: Fraction of values above μ
10) **Number of peaks**: Local maxima count (prominence-based)
11) **RMS**: Root mean square energy
12) **IQR**: Interquartile range (robust spread)
13) **Signal energy**: Sum of squared values

All features are computed via vectorized NumPy/SciPy operations in $\sim$4.85 ms per window. Per-(feature-row, sensor) normalization is applied to account for the vastly different scales of each statistic.

B. Model Architectures

1) Bidirectional LSTM with Attention: Our LSTM architecture uses 2 bidirectional layers with hidden dimension 256 and LayerNorm between layers. Rather than using the last hidden state, we apply an attention mechanism over all timesteps:

$$\alpha_t = \frac{\exp(\mathbf{w}^\top \mathbf{h}_t)}{\sum_{t'} \exp(\mathbf{w}^\top \mathbf{h}_{t'})}, \quad \mathbf{c} = \sum_t \alpha_t \mathbf{h}_t \tag{1}$$

where $\mathbf{h}_t$ is the bidirectional hidden state at timestep t and $\mathbf{w}$ is a learned attention vector. The context vector $\mathbf{c}$ feeds into a classification head (Linear $\rightarrow$ GELU $\rightarrow$ Dropout $\rightarrow$ Linear). Total parameters: 2.30M (classification), 2.24M (forecasting).

2) DeepONet: DeepONet [4] learns operators mapping input functions to output functions through branch and trunk networks. Our implementation provides three interchangeable branch architectures for compact feature matrices (window size $\leq$ 30): a flat feedforward branch (`mlp`, used for the reported results) that treats the flattened (14×27) input as a single 378-dimensional vector, a `conv1d` branch that convolves across the 27-sensor axis treating the 14 descriptors as input channels, and an `attention` branch that treats each sensor as a token with a 14-dimensional embedding and applies self-attention. The MLP branch avoids spurious spatial assumptions among heterogeneous statistical descriptors (mean, kurtosis, slope, energy). For classification, the trunk network and the inner-product output mechanism are *entirely disabled* (`self.trunk = None`); the branch functions purely as a feature encoder, and a classification head (Linear $\rightarrow$ GELU $\rightarrow$ Dropout $\rightarrow$ Linear) maps the branch embedding directly to class logits. For the forecasting and anomaly tasks, the full DeepONet architecture is used: the trunk encodes query positions and the output is the bilinear product $\mathbf{b}^\top \mathbf{t} + \text{bias}$. Total parameters: 209K (classification), 210K (forecasting).

3) PatchTST: PatchTST [5] segments multivariate time series into patches and applies a channel-independent Transformer encoder. We configure 3 layers, d_{model}=256, d_{ff}=512, 8 attention heads, and patch length adapted to input size (4 for feature matrices, 16 for raw sequences). Total parameters: 433K (classification), 402K (forecasting).

4) Random Forest: As a non-sequential baseline for 3W classification, we train a scikit-learn `RandomForestClassifier` on the flattened $(14 \times 27) \rightarrow$ 378-dimensional feature vector. Unlike the recurrent, operator-learning, and attention-based architectures, Random Forest treats the feature matrix as an unstructured tabular input—testing whether an ensemble of decision trees can extract discriminative patterns from the precomputed statistical descriptors without exploiting their spatial arrangement. We use 500 estimators, Gini impurity, balanced class weights (inverse frequency), and evaluate via `StratifiedGroupKFold` 5-fold CV. Total parameters: $\sim$2.4M leaf nodes across the ensemble (non-gradient-based).

5) Temporal Convolutional Network (TCN): For Ganymede forecasting, we employ a Temporal Convolutional Network (TCN) that applies stacked dilated causal 1D convolutions with residual connections. Dilation factors grow exponentially ($d_i = 2^i$), giving the network a receptive field that covers the full 90-timestep input window with only $\mathcal{O}(\log T)$ layers. Each residual block consists of two dilated convolutions with weight normalization, GELU activations, and spatial dropout. A final linear projection maps the last temporal output to multi-step forecasts at horizons $h \in \{7, 14, 30, 90\}$. Unlike RNNs, the TCN processes the entire sequence in parallel, enabling efficient training while preserving strict causal ordering. We configure 128 channels, 6 layers (dilation $1, 2, \ldots, 32$, yielding a receptive field of 253 timesteps), kernel size 3, dropout 0.2, learning rate 10^{-3} with weight decay 10^{-4}.

6) FKM-AD (Fourier-KAN Mamba Anomaly Detector): FKM-AD [12] combines Fourier-basis Kolmogorov–Arnold projections with selective state-space modeling (Mamba) for time-series classification. The pipeline processes the (14×27) feature matrix—or raw (720×27) windows—through: (1) a FourierKANProjection layer that projects each input channel to d_{model}=256 via learnable Fourier basis functions (15 frequencies, rank 32); (2) LayerNorm; (3) two stacked Mamba SSM layers (d_{state}=32, d_{conv}=4, expand= 2) with residual connections; (4) a GatedSharpeningTemperature module that adaptively sharpens per-timestep logits; and (5) AttentionPooling that produces a fixed-length representation. A linear classification head maps the pooled output to class logits. Dropout (0.39) is applied throughout.

We evaluate FKM-AD in two data paths: *feature*, operating on the pre-extracted (14×27) statistical matrix, and *raw*, operating directly on the (720×27) sensor windows with per-instance z-score normalization. This dual-path evaluation quantifies the contribution of feature engineering vs. the SSM's native sequence-modeling capability. Total parameters: 0.26M.

7) MambaSL (Single-Layer Mamba with Time-Variant SSM): MambaSL [13] is a compact state-space model that employs a single Mamba layer with time-variant SSM parameters and multi-head adaptive attention pooling. Unlike FKM-AD's two-stage Fourier-KAN projection followed by stacked Mamba blocks, MambaSL uses a direct embedding projection into the SSM layer without Fourier basis pre-conditioning. The attention pooling mechanism aggregates temporal features into a fixed-length vector, which feeds a linear classification head. Total parameters: 0.43M.

We evaluate MambaSL on the same dual data paths as FKM-AD: the *feature* path operates on pre-extracted (14×27) statistical matrices, and the *raw* path operates directly on (720×27) sensor windows with per-instance z-score normalization. MambaSL hyperparameters are optimized via 30-trial Optuna HPO on the feature path (D018).

8) ConvTimeNet (Deformable Patch Conv): ConvTimeNet [14] is a deep hierarchical fully convolutional model for multivariate time-series analysis. Its key innovations are: (1) a deformable patch embedding that predicts per-patch spatial offsets, allowing the model to adapt receptive fields to lo-

TABLE II: Training hyperparameters (manual, pre-HPO).

Parameter	LSTM	DeepONet	RF	PatchTST	FKM-AD
Learning rate	1e-3	5e-4	—	1e-4	1e-3
Weight decay	1e-4	1e-4	—	1e-2	1e-4
Batch size	64	64	—	64	64
LR scheduler	OneCycle	Cosine	—	OneCycle	Cosine
Max epochs	100	100	—	100	100
Dropout	0.3	0.1	—	0.1	0.2
n_estimators	—	—	500	—	—
class_weight	—	—	balanced	—	—
d_model	—	—	—	—	128
d_state	—	—	—	—	16

cal signal dynamics rather than using fixed-stride windows; (2) multi-scale depthwise convolution blocks that capture features at multiple temporal resolutions; and (3) structural re-parameterization at evaluation time, which folds auxiliary branches into a single Conv1d operator, eliminating any inference overhead. All operations are pure `Conv1d`/`Conv2d` (CPU-native, no CUDA dependencies), making ConvTimeNet deployable without GPU infrastructure. Total parameters: ∼0.27M (task-specific head included).

We evaluate ConvTimeNet on both data paths: the feature path (14×27 statistical matrix) and the raw path (720×27 time-step windows), consistent with the dual-path protocol used for FKM-AD and MambaSL. Hyperparameters are optimized via 30-trial Optuna HPO on the feature path, consistent with the other HPO-tuned models; the raw path uses default hyperparameters.

9) Foundation Models: **TiRex** [8]: We extract frozen xLSTM embeddings (6,144-dimensional) for each 3W window and train a Random Forest classifier (500 trees, balanced class weights) on the embedding space. Embeddings are extracted once to a memory-mapped file (4.8 GB float32) to avoid GPU memory issues.

Chronos [6]: Amazon's zero-shot probabilistic forecaster generates quantile predictions via tokenized time-series language modeling. Used for Ganymede forecasting.

TimesFM [7]: Google's patched-decoder foundation model provides zero-shot point forecasts. Used for Ganymede forecasting.

C. Training Configuration

All trained models use AdamW optimizer with gradient clipping at 1.0 and early stopping (patience = 20 epochs). Model-specific hyperparameters are listed in Table II. Balanced class weights (inverse frequency) are used for 3W classification to address imbalance.

V. Evaluation Protocol

We adopt a nested evaluation design (Fig. 1) to produce unbiased test metrics while also providing variance estimates for model stability assessment.

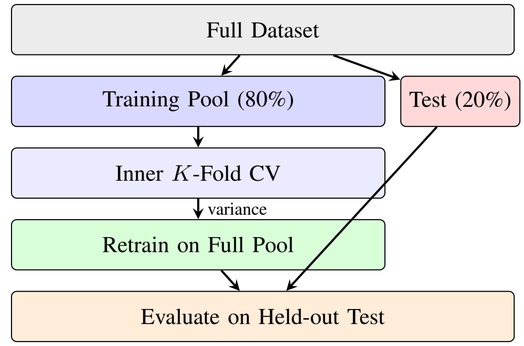

Fig. 1: Nested evaluation protocol. The outer split creates a held-out test set; inner CV provides variance estimates; a fresh model retrained on the full training pool is evaluated on the test set.

A. Outer Holdout Split

3W (Classification): We use a stratified group split (80/20) where groups are defined by well instance ID. This ensures that (a) no windows from the same well instance appear in both training and test sets (preventing data leakage), and (b) class proportions are preserved across the split. The resulting split yields 167,458 training and 41,515 test samples with 1,261 and 307 groups, respectively, and zero group overlap.

Ganymede (Forecasting): A temporal split reserves the most recent 20% of each well's history as the test set, preserving temporal ordering and preventing future information leakage.

B. Inner Cross-Validation

Within the 80% training pool, we run inner CV to estimate model variance:

- **3W**: 5-fold Stratified Group K-Fold, maintaining group integrity and class balance within each fold.
- **Ganymede**: 3-fold Expanding Window CV with minimum training ratio 0.5, respecting temporal ordering.

C. Final Model and Test Evaluation

After inner CV, a fresh model is trained on the entire 80% training pool and evaluated on the held-out 20% test set. The test metrics constitute our primary results; inner CV means ± standard deviations are reported as diagnostics.

D. Metrics

Classification (3W): Accuracy, macro F_1 (unweighted average across classes), weighted F_1 (class-size weighted), and AUC-PR (area under the precision-recall curve, macro-averaged). We also report the Event Detection Rate (EDR), the fraction of classes with non-zero F_1.

Forecasting (Ganymede): MAE, RMSE, R^2, and R^2_{prod}—the coefficient of determination computed only on productive windows (target > 0.01). R^2_{prod} isolates forecasting quality from shutdown prediction, providing a more meaningful measure for production planning.

VI. Results

A. 3W Fault Classification

Table III presents the primary results on the held-out test set (n=41,515). All eleven model configurations are evaluated on identical test samples. DeepONet and LSTM achieve near-identical accuracy (∼96.8%), while PatchTST reaches 96.5% after Optuna HPO (30 trials, improved from 95.7% with manual tuning). Notably, Random Forest achieves 96.58% accuracy (0.966, macro F_1=0.964, AUC-PR=0.985) on the flattened 378-dimensional feature vector, demonstrating that the statistical feature extraction pipeline captures discriminative information accessible even to non-sequential, tabular learners. FKM-AD on features achieves 96.7% accuracy (F_1m=0.961), competitive with the top models, while the raw-window variant reaches 94.9%—a 1.8 percentage point gap that quantifies the value of the feature extraction pipeline for state-space models. Inner CV estimates are within 2% of test performance for all trained models, indicating stable generalization.

MambaSL [13] provides a second dual-path evaluation on the same classification task. The feature path achieves 96.6% accuracy (F_1m=0.961, AUC-PR=0.927, CV=95.2%), ranking fourth behind DeepONet, LSTM, and FKM-AD (feat.). The raw-window variant reaches 91.9% (F_1m=0.887, CV=93.0%)—a 4.7 pp feature-vs-raw gap that is wider than FKM-AD's 1.8 pp gap. This gap suggests that MambaSL's single-layer time-variant SSM design, which lacks FKM-AD's explicit Fourier-basis frequency decomposition, benefits more from pre-extracted statistical features when processing the high-dimensional 27-sensor input space.

ConvTimeNet [14] is evaluated on both data paths. The feature path achieves 96.6% accuracy (F_1m=0.962, AUC-PR=0.930, CV=95.5% ± 1.5%), ranking third among all configurations after Random Forest and FKM-AD (feat.). The raw-window path achieves 15.8% accuracy (F_1m=0.027, AUC-PR=0.100)—an 80.8 pp gap, the largest among the three dual-path models (FKM-AD: 1.8 pp, MambaSL: 4.7 pp). This dramatic degradation indicates that ConvTimeNet's deformable patch embedding and structural re-parameterization are optimized for compact feature representations rather than raw 720-step sequences, making it the architecture most dependent on the feature extraction pipeline.

TiRex, using frozen embeddings without any neural fine-tuning, achieves 91.2% accuracy—a remarkably strong zero-shot baseline. Notably, it achieves the *highest* AUC-PR (0.938), suggesting well-calibrated class probability estimates from the Random Forest on the 6,144-dimensional embedding space.

1) Per-Class Analysis: Fig. 2 shows per-class recall on the held-out test set for all eight trained models plus TiRex. Classes 5 (flow instability), 6 (rapid productivity loss), and 9 (hydrate formation) achieve >99% recall across most trained models. Random Forest achieves the highest recall on class 2 (incipient BSW increase, 93.2%), the rarest class, outperforming all neural architectures on this challenging category. FKM-AD (features) achieves 100% recall on class 6 and 99.8% on class 9, closely tracking the top models, but shows reduced recall on class 2 (89.9%) and class 4 (92.2%). The raw-window variant follows a similar pattern with somewhat lower recall, confirming that the SSM benefits from pre-extracted features, though the gap narrows substantially after hyperparameter optimization. PatchTST shows reduced recall on class 8 (slow restriction, 87%), while TiRex shows the largest drops on class 2 (67.8%) and class 4 (73.5%), indicating that frozen embeddings lose fine-grained discriminative information for rare event types. MambaSL (feat.) shows strong recall across all classes (>89% for all 10 classes), with particularly high performance on classes 5 (99.4%), 6 (99.8%), and 9 (99.9%), and reduced recall on class 2 (89.7%) compared to top models.

2) Confusion Matrix: Table IV presents the confusion matrix for DeepONet (best overall) on the held-out test set. The dominant off-diagonal errors are: class 0→8 (396 normal windows misclassified as slow restriction), class 1→0 (275

TABLE III: 3W fault classification results. **Test** = held-out 20% (n=41,515); **CV** = inner 5-fold mean±std on the 80% training pool. Best test values in bold.

	Held-out Test				CV A
Model	Acc.	F_1m	F_1w	AUC-PR	mean±
DeepONet	**.968**	**.964**	**.969**	.934	.956±
LSTM	.968	.963	.968	.933	.949±
PatchTST†	.965	.962	.965	.931	.964±
RF	.966	.964	.966	.985	.979±
FKM-AD (feat.)	.967	.961	.967	.929	.960±
FKM-AD (raw)	.949	.938	.949	.888	.936±
ConvTimeNet (feat.)†	.966	.962	.966	.930	.955±
ConvTimeNet (raw)	.158	.027	.043	.100	.631±
MambaSL (feat.)†	.966	.961	.966	.927	.952±
MambaSL (raw)†	.919	.887	.920	.807	.930±
TiRex (FM)	.912	.895	.911	**.938**	.899±
Majority	*.174*	*.101*	—	—	—

†After Optuna HPO (30 trials). DeepONet: 33 trials. FKM-AD (feat.): 30 trials; FKM-AD (raw): baseli

†MambaSL, ConvTimeNet hyperparameters tuned via 30-trial Optuna HPO on feature path. ConvTimeN

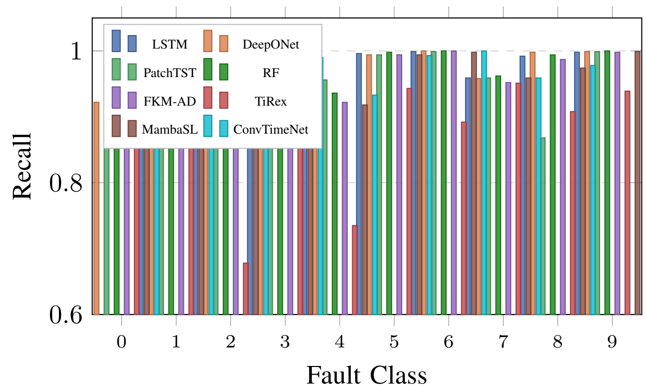

Fig. 2: Per-class recall on the 3W held-out test set for all eight trained models plus TiRex. Classes: 0=Normal, 1=Abrupt BSW↑, 2=Incipient BSW↑, 3=Spurious closure, 4=Severe slugging, 5=Flow instability, 6=Productivity loss, 7=Quick restriction, 8=Slow restriction, 9=Hydrate.

TABLE IV: Confusion matrix: DeepONet on 3W held-out test set (n=41,515). Rows = true, columns = predicted. Only cells $\geq$ 10 shown for clarity; blank cells are $<$ 10.

	0	1	2	3	4	5	6	7	8	9
0	**6049**				105				396	
1	275	**4592**			40					
2			**355**				11		24	
3		37		**2659**						
4	84				**1817**					
5						**7215**	39			
6							**3159**			
7	201				17			**5432**	17	
8									**3723**	
9										**5205**

TABLE V: Ganymede forecasting (held-out test, multi-well, MMSCF). Best values per column in bold. FM = foundation model (zero-shot, no training).

Model	**Metric**	h=7	h=14	h=30	h=90
TimesFM[FM]	R^2_{prod}	**.826**	.763	.656	.413
	MAE	.740	.868	.949	1.02
TiRex[FM]	R^2_{prod}	.822	**.775**	**.699**	**.565**
	MAE	**.658**	**.749**	**.804**	**.831**
Chronos[FM]	R^2_{prod}	.734	.668	.584	.361
	MAE	.739	.835	.886	.995
PatchTST	R^2_{prod}	.556	.466	.211	−.382
	MAE	1.77	1.98	2.14	2.48
LSTM	R^2_{prod}	.195	.534	.303	.020
	MAE	1.77	1.69	1.84	2.08
DeepONet	R^2_{prod}	−40.1	−4.35	−5.01	−4.32
	MAE	9.53	4.12	4.51	3.84
TCN	R^2_{prod}	−4.51	.108	.060	−.044
	MAE	3.98	2.37	2.62	2.34

abrupt BSW misclassified as normal), and class 7→0 (201 quick restriction misclassified as normal). These confusions are physically plausible—normal operation shares sensor signatures with early-onset restriction and BSW changes.

B. Ganymede Gas Production Forecasting

Table V presents multi-horizon forecasting results on the Ganymede multi-well setting, with gas production measured in MMSCF (million standard cubic feet). All three foundation models achieve positive R^2_{prod} across all four horizons, with TiRex leading at longer horizons (h=30: 0.70, h=90: 0.57) and TimesFM at the shortest (h=7: 0.83). Among trained models, PatchTST and LSTM achieve positive R^2_{prod} at most horizons, while DeepONet yields consistently negative R^2. TCN achieves marginally positive R^2_{prod} at h=14 (0.11) and h=30 (0.06) but yields strongly negative values at h=7 ($-$4.51) and h=90 ($-$0.04), ranking below both PatchTST and LSTM.

Fig. 3 visualizes R^2_{prod} across horizons. Foundation models maintain positive R^2_{prod} at all horizons, with TiRex showing the most graceful degradation (0.82→0.57). Among trained models, PatchTST degrades from 0.56 (h=7) to $-$0.38 (h=90). LSTM shows an unusual pattern: low R^2_{prod} at h=7 (0.20)

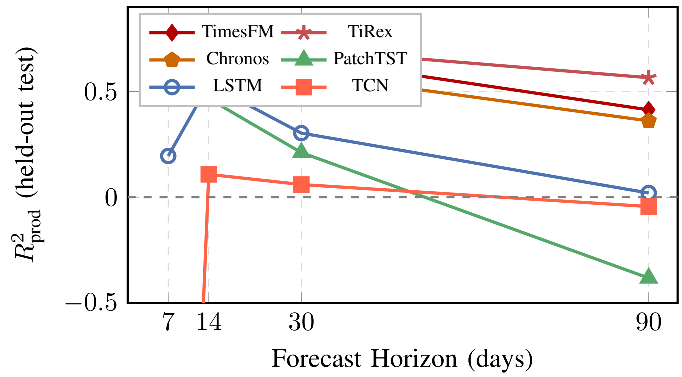

Fig. 3: R^2_{prod} vs. forecast horizon (Ganymede, held-out test, multi-well). Foundation models (top 3 curves) consistently outperform trained models. DeepONet omitted ($R^2 < -5$). TCN omitted from y-axis range due to R^2_{prod}=$-$4.51 at h=7. Dashed line: R^2=0 (naive baseline).

but peaking at h=14 (0.53), possibly reflecting how its recurrent architecture captures medium-term trends while struggling with short-term variability. DeepONet consistently fails ($R^2 < -4$), confirming that its operator-learning formulation is poorly suited for autoregressive production forecasting. TCN, despite its causal convolutional architecture designed for temporal modeling, underperforms both PatchTST and LSTM: its strongly negative R^2_{prod} at h=7 ($-$4.51) suggests overfitting or poor generalization on the shortest horizon, while its marginal results at h=14 and h=30 ($R^2_{\text{prod}} \approx 0.1$) indicate that dilated causal convolutions alone do not confer sufficient inductive bias for this multi-well forecasting setting.

VII. Discussion

A. Classification: Trained Models vs. Foundation Models

All four trained classification models achieve strong performance (>96% accuracy after HPO), confirming that the statistical feature extraction pipeline captures the essential fault signatures. DeepONet is the most parameter-efficient (209K) while marginally leading in accuracy (96.85%). Architecturally, this result is notable: the classification variant disables the trunk network entirely, bypassing the inner-product operator mechanism that defines DeepONet. The branch network—originally designed as a "function encoder" for operator learning—transfers effectively as a standalone fault-signature encoder, and the classification head maps the branch embedding directly to class logits. To our knowledge, this is the first application of DeepONet to multiclass time-series classification; the finding that the branch-as-encoder pattern suffices without the trunk suggests that DeepONet's function-encoding inductive bias, not the operator-output mechanism, drives its classification strength. Optuna hyperparameter optimization (30 trials per model; 33 for DeepONet) improved PatchTST from 95.7% to 96.5% by finding a deeper, narrower architecture (4 layers, d_{model}=128), while LSTM and DeepONet showed no improvement—confirming that manual tuning was near-optimal for these architectures.

Random Forest's success at 96.58% accuracy on the flattened 378-dimensional feature vector is the most illuminating result for the inductive bias debate. Unlike the neural models that process the (14×27) feature matrix as a structured sequence, Random Forest treats the same features as unordered tabular input—yet achieves comparable performance. This demonstrates that the *feature extraction pipeline itself* captures the discriminative structure: the 14 statistical descriptors per sensor encode temporal dynamics (slope, mean absolute change, peaks) in a form that is accessible to any competent learner, regardless of whether it models spatial or sequential relationships among features. The inductive bias argument is therefore more nuanced than a binary structured-vs-flat distinction: when feature engineering explicitly captures temporal structure, even non-sequential models can exploit the resulting representation.

FKM-AD provides further evidence from the state-space model family. On the pre-extracted feature matrix, FKM-AD achieves 96.7% accuracy (CV=96.0%) with only 0.26M parameters—competitive with architectures 2–9× larger—while the raw-window variant reaches 94.9% (CV=93.6%). This 1.8 pp gap between feature and raw paths quantifies the benefit of statistical feature extraction for SSMs: even Mamba's selective state-space mechanism, designed for long-range sequence modeling, performs better when temporal dynamics are pre-encoded as statistical descriptors, though the gap is smaller than initially expected. The dual-path result complements Random Forest's success—both demonstrate that the feature extraction pipeline, not the downstream architecture's inductive bias, is the primary driver of classification performance.

MambaSL [13] provides additional evidence that widens the picture. With a single-layer time-variant SSM and multi-head adaptive attention pooling (0.43M parameters), MambaSL achieves 96.6% accuracy on features (CV=95.2%) but 91.9% on raw windows (CV=93.0%)—a 4.7 pp feature-vs-raw gap. This exceeds FKM-AD's 1.8 pp gap, suggesting that MambaSL's architectural choices—in particular, the absence of Fourier-basis frequency pre-conditioning—benefit more from pre-extracted statistical features. The pattern is visible in the class recall analysis: MambaSL (feat.) achieves >89% recall on every class, while the raw variant drops below 84% on class 0, class 3, and class 4. Taken together, FKM-AD and MambaSL establish that the feature extraction benefit is a structural property of state-space models applied to this domain: the magnitude of the benefit varies with the SSM's frequency-handling capability, but it is positive in both cases.

ConvTimeNet [14] introduces a third inductive bias family—pure convolution—into the dual-path analysis. On the feature path, its deformable patch embedding with per-patch offset prediction and multi-scale depthwise convolution blocks with structural re-parameterization achieve 96.6% accuracy (rank 3.8 of 11 configurations), competitive with both attention-based and SSM-based architectures. On the raw path (720 × 27), however, ConvTimeNet collapses to 15.8% accuracy—an 80.8 pp gap that dwarfs FKM-AD's 1.8 pp and MambaSL's 4.7 pp gaps. This result reveals that ConvTimeNet's patch-based design, while powerful for compact feature matrices, cannot effectively learn from raw sensor sequences at this scale. The three dual-path gaps (1.8, 4.7, 80.8 pp) form a spectrum of feature extraction dependence that correlates with architectural specificity: FKM-AD's Fourier-KAN frequency decomposition provides built-in signal processing that partially compensates for the absence of pre-extracted features; MambaSL's simpler SSM requires more feature engineering support; and ConvTimeNet's deformable convolutions, designed for compact inputs, fail entirely on raw sequences. This gradient further strengthens the thesis that feature extraction is the primary performance driver on this dataset.

TiRex provides a strong zero-shot baseline (91.2%) that would be adequate for initial deployment without labeled data. However, its 5.7 percentage point gap from trained models, concentrated in rare classes (67% recall on class 2), suggests that domain-specific fine-tuning or at least a small amount of labeled data is needed for safety-critical applications.

B. Forecasting: Foundation Models Dominate

The Ganymede results reveal a striking finding: all three foundation models substantially outperform all trained models at every forecast horizon. TiRex achieves the best overall performance (R^2_{prod}=0.82 at h=7, 0.57 at h=90), followed closely by TimesFM (R^2_{prod}=0.83 at h=7) and Chronos (R^2_{prod}=0.73 at h=7). The best trained model, PatchTST, peaks at R^2_{prod}=0.56 and degrades to negative R^2 at h=90.

This performance gap likely reflects the FMs' pre-training on diverse time-series corpora: production data shares temporal patterns (trends, seasonality, autocorrelation) with the training distributions, enabling effective zero-shot transfer. In contrast, the trained models must learn these patterns from ∼25,000 samples with considerable well heterogeneity, insufficient for capturing the full dynamics.

Notably, the contrast between Random Forest's success on 3W classification and the forecasting results highlights a task-dependent role for feature engineering: on 3W, the statistical descriptors explicitly encode temporal dynamics in a tabular-friendly form, enabling non-sequential models; on Ganymede, no such feature extraction is applied, and models must learn temporal relationships directly from raw sequences—a setting where sequential inductive biases were expected to confer advantage. However, TCN's poor performance ($R^2_{\text{prod}} \leq 0.11$ across horizons) demonstrates that causal convolutions alone are insufficient: the attention-based PatchTST (R^2_{prod}=0.56 at h=7) substantially outperforms the purely convolutional TCN, suggesting that global attention over patched subsequences provides a more effective inductive bias than local dilated convolutions for multi-well production forecasting.

C. Evaluation Protocol Validation

The close agreement between inner CV estimates and held-out test performance (within 2% for 3W) validates both the stratified-group splitting strategy and the overall protocol. The fact that test accuracy slightly *exceeds* CV accuracy (96.8% vs. 94.9–95.5%) is expected: the retrained model benefits from the full 80% training pool versus the ∼64% available in each inner CV fold.

D. Statistical Significance

We apply the Friedman test with Nemenyi post-hoc analysis to the inner CV fold-level metrics (5 folds for 3W, 3 folds for Ganymede). For 3W classification with eleven model configurations (including dual paths for FKM-AD, MambaSL, and ConvTimeNet), the Friedman test is significant across all metrics: accuracy (χ^2=43.5, $p < 0.001$), and similarly for macro F_1 and AUC-PR. Average accuracy ranks place Random Forest first (1.0), FKM-AD features second (3.0), ConvTimeNet features third (3.8), PatchTST fourth (4.0), DeepONet fifth (4.8), LSTM sixth (5.6), MambaSL features seventh (5.8), FKM-AD raw eighth (8.6), MambaSL raw ninth (9.2), ConvTimeNet raw tenth (9.6), and TiRex eleventh (10.6). With Nemenyi CD = 6.75 at α=0.05, significant pairwise differences are found between: ConvTimeNet features and TiRex (rank diff= 6.8); ConvTimeNet raw and Random Forest (rank diff= 8.6); FKM-AD features and TiRex (rank diff= 7.6); FKM-AD raw and Random Forest (rank diff= 7.6); MambaSL raw and Random Forest (rank diff= 8.2); and Random Forest and TiRex (rank diff= 9.6). These results confirm that all feature-path models are statistically indistinguishable from each other, while raw-window variants and TiRex form a separate lower tier—with ConvTimeNet raw joining TiRex at the bottom, reflecting its extreme dependence on the feature extraction pipeline.

For Ganymede forecasting with seven models (including TCN), MAE reaches significance at h=7 (p=0.009), h=14 (p=0.017), and h=30 (p=0.037), with limited statistical power from 3 inner CV folds. Average MAE ranks at h=7 place TiRex first (1.0), Chronos second (2.3), TimesFM third (3.0), LSTM fourth (3.7), PatchTST fifth (5.0), DeepONet sixth (6.3), and TCN last (6.7)—confirming that TCN's causal convolutional architecture does not confer a competitive advantage on this multi-well forecasting task.

E. Limitations

This study has several limitations: (i) hyperparameter optimization was performed for the 3W classification task but not for the trained models on Ganymede forecasting, where the large gap between FMs and trained models may partly reflect suboptimal tuning; and (ii) no interpretability analysis (e.g., attention visualization, SHAP values) has been conducted.

VIII. Conclusion and Future Work

We benchmarked ten architectures on offshore fault classification and gas production forecasting using a rigorous nested evaluation protocol. Key findings:

1) **Statistical features enable efficient classification**: Compressing 720-step windows into 14-descriptor matrices yields >96% accuracy on 3W fault detection with all four trained models (after HPO), while reducing training time by ∼50×.
2) **Feature engineering captures temporal structure**: Random Forest achieves 96.58% accuracy on the flattened 378-dimensional feature vector, demonstrating that the statistical extraction pipeline encodes discriminative temporal dynamics in a form accessible to non-sequential, tabular learners—the inductive bias advantage is more nuanced than a simple structured-vs-flat distinction.
3) **SSMs benefit from feature extraction**: FKM-AD (Fourier-KAN Mamba) achieves 96.7% accuracy on pre-extracted features but reaches 94.9% on raw windows—a 1.8 pp gap demonstrating that selective state-space models, designed for long-range sequence modeling, still perform better when temporal dynamics are pre-encoded as statistical descriptors, though the gap narrows with proper training stabilization (instance normalization, NaN-safe training).
4) **MambaSL widens the feature-extraction case for SSMs**: MambaSL [13] (single-layer time-variant SSM) achieves 96.6% on pre-extracted features but 91.9% on raw windows after HPO—a 4.7 pp gap larger than FKM-AD's 1.8 pp. The magnitude of the feature-extraction benefit varies with the SSM's frequency-handling capability, but it is positive for both architectures, strengthening the argument that statistical feature extraction is the primary performance driver for SSMs on this task.
5) **ConvTimeNet reveals extreme feature extraction dependence**: ConvTimeNet [14] (pure-convolution, deformable patches with structural re-parameterization) achieves 96.6% accuracy on pre-extracted features (rank 3.8 of 11 configurations) but collapses to 15.8% on raw windows—an 80.8 pp gap, the largest among three dual-path models (FKM-AD: 1.8 pp, MambaSL: 4.7 pp). This spectrum of feature extraction dependence demonstrates that architectural inductive biases determine raw-signal performance while feature extraction equalizes all architectures to the 96.5–96.7% accuracy band.
6) **DeepONet leads classification, fails forecasting**: The operator-learning framework achieves the best classification accuracy (96.85%) with the fewest parameters (209K), using only the branch network as a feature encoder—the trunk and inner-product mechanism are entirely bypassed for classification. This demonstrates that DeepONet's function-encoding inductive bias transfers to fault signature encoding without the operator-output mechanism. However, the same architecture produces negative R^2 on all forecasting horizons when the trunk is re-enabled, indicating that the operator formulation does not generalize to autoregressive production forecasting.
7) **PatchTST is the most versatile trained model**: Achieves R^2_{prod}=0.56 at the 7-day horizon (best among trained models) and 96.5% classification accuracy after HPO.

8) **Foundation models dominate forecasting**: TimesFM, TiRex, and Chronos all achieve $R^2_{\text{prod}} > 0.7$ at h=7 without any task-specific training, exceeding the best trained model by 24–48%. TiRex maintains the strongest performance at long horizons (R^2_{prod}=0.57 at h=90).
9) **FMs provide competitive classification baselines**: TiRex achieves 91.2% zero-shot classification with the highest AUC-PR (0.938), demonstrating the versatility of pre-trained time-series representations for offshore monitoring.
10) **TCN underperforms on Ganymede forecasting**: Despite its causal convolutional architecture and receptive field covering the full input window, TCN achieves $R^2_{\text{prod}} \leq 0.11$ across all horizons—substantially below PatchTST (0.56 at h=7) and LSTM (0.53 at h=14)—indicating that dilated causal convolutions alone are insufficient for multi-well production forecasting without task-specific feature engineering or attention mechanisms.

Future work includes extending trained models to the forecasting task, model interpretability analysis, and evaluation of domain-adapted foundation models.

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